

ct > src > App.jsx > App

```
1  import { useState } from "react";
2  function App() {
3      const [weight, setWeight] = useState("");
4      const [height, setHeight] = useState("");
5      const [bmi, setBmi] = useState(null);
6      const [category, setCategory] = useState("");
7      const calculateBMI = () => {
8          if (!weight || !height) {
9              setCategory("Please enter weight and height");
10             setBmi(null);
11             return;
12         }
13         const heightInMeter = height / 100;
14         const bmiValue = weight / (heightInMeter * heightInMeter);
15         setBmi(bmiValue.toFixed(2));
16         if (bmiValue < 18.5) {
17             setCategory("Underweight");
18         } else if (bmiValue < 25) {
19             setCategory("Normal Weight");
20         } else if (bmiValue < 30) {
21             setCategory("Overweight");
22         } else {
23             setCategory("Obesity");
24         }
25     };
26 }
```

```

const reset = () => {
  setWeight("");
  setHeight("");
  setBmi(null);
  setCategory("");
};

return (
  <div>
    <h1>BMI Calculator</h1>
    <label>Weight (kg): </label>
    <input
      type="number"
      value={weight}
      onChange={(e) => setWeight(e.target.value)}
    />
    <br />
    <label>Height (cm): </label>
    <input
      type="number"
      value={height}
      onChange={(e) => setHeight(e.target.value)}
    />
    <br />
    <br />
    <button onClick={calculateBMI}>Calculate BMI</button>

    <button onClick={reset}>Reset</button>
  </div>
)

```

```

    {bmi && (
      <div>
        <h2>Your BMI: {bmi}</h2>
        <h3>Category: {category}</h3>
      </div>
    )}
  </div>
);
}

export default App;

```

2)Age calculator

```
App.jsx M • localhost:5173 × react ×
react > src > App.jsx > App > calculateAge
1  import { useState } from "react";
2  function App() {
3      const [dob, setDob] = useState("");
4      const [age, setAge] = useState(null);
5      const calculateAge = () => {
6          if (!dob) {
7              alert("Please select your Date of Birth");
8              return;
9          }
10         const birthDate = new Date(dob);
11         const today = new Date();
12         let years = today.getFullYear() - birthDate.getFullYear();
13         let months = today.getMonth() - birthDate.getMonth();
14         let days = today.getDate() - birthDate.getDate();
15         if (days < 0) {
16             months--;
17             const lastMonth = new Date(
18                 today.getFullYear(),
19                 today.getMonth(),
20                 0
21             ).getDate();
22             days += lastMonth;
23         }
24         if (months < 0) {
25             years--;
26             months += 12;
27         }
28         setAge({ years, months, days });
29     };
30     const reset = () => {
31         setDob("");
32         setAge(null);
33     };
34 }
```


3)Calci

```
App.jsx M  localhost:5173  react X
react > src > App.jsx > App
1  import { useState } from "react";
2
3  function App() {
4      const [num1, setNum1] = useState("");
5      const [num2, setNum2] = useState("");
6      const [result, setResult] = useState("");
7
8      const add = () => {
9          setResult(Number(num1) + Number(num2));
10     };
11
12     const subtract = () => {
13         setResult(Number(num1) - Number(num2));
14     };
15
16     const multiply = () => {
17         setResult(Number(num1) * Number(num2));
18     };
19
20     const divide = () => {
21         if (Number(num2) === 0) {
22             setResult("Cannot divide by zero");
23         } else {
24             setResult(Number(num1) / Number(num2));
25         }
26     };
27
28     const reset = () => {
29         setNum1("");
30         setNum2("");
31         setResult("");
32     };
33
```

```

return (
  <div style={{ textAlign: "center", marginTop: "50px" }}>
    <h1>Simple Calculator</h1>

    <input
      type="number"
      placeholder="Enter First Number"
      value={num1}
      onChange={(e) => setNum1(e.target.value)}
    />

    <br /><br />

    <input
      type="number"
      placeholder="Enter Second Number"
      value={num2}
      onChange={(e) => setNum2(e.target.value)}
    />
    <br /><br />
    <button onClick={add}>+</button>
    <button onClick={subtract}>-</button>
    <button onClick={multiply}>*</button>
    <button onClick={divide}>/</button>
    <button onClick={reset}>Reset</button>

    <h2>Result: {result}</h2>
  </div>
);
}

export default App;

```

4)

To do list

react > src > App.jsx > App > toggleComplete

```
1  import { useState } from "react";
2  function App() {
3    const [task, setTask] = useState("");
4    const [tasks, setTasks] = useState([]);
5    const addTask = () => {
6      if (task.trim() === "") return;
7      setTasks([
8        ...tasks,
9        {
10         id: Date.now(),
11         text: task,
12         completed: false,
13       },
14     ]);
15
16     setTask("");
17   };
18   const deleteTask = (id) => {
19     setTasks(tasks.filter((t) => t.id !== id));
20   };
21   const toggleComplete = (id) => {
22     setTasks(
23       tasks.map((t) =>
24         t.id === id
25         ? { ...t, completed: !t.completed }
26         : t
27       )
28     );
29   };
30   const clearCompleted = () => {
31     setTasks(tasks.filter((t) => !t.completed));
32   };
```

```

31   return (
32     <div style={{ padding: "20px", textAlign: "center" }}>
33       <h1>To-Do List</h1>
34       <input
35         type="text"
36         placeholder="Enter a task"
37         value={task}
38         onChange={(e) => setTask(e.target.value)}
39       />
40
41       <button onClick={addTask}>Add Task</button>
42
43       <ul style={{ listStyle: "none", padding: 0 }}>
44         {tasks.map((t) => (
45           <li key={t.id} style={{ margin: "10px" }}>
46             <span
47               onClick={() => toggleComplete(t.id)}
48               style={{
49                 cursor: "pointer",
50                 textDecoration: t.completed
51                   ? "line-through"
52                   : "none",
53                 marginRight: "10px",
54               }}
55             >
56               {t.text}
57             </span>
58
59             <button onClick={() => deleteTask(t.id)}>
60               Delete
61             </button>
62           </li>
63         ))}
64       </ul>
65
66       {tasks.length > 0 && (
67         <button onClick={clearCompleted}>
68           Clear Completed Tasks
69         </button>
70       )}
71     </div>
72   );
73 }
74
75 export default App;

```

5)

Currency converter


```

function App() {
  const [amount, setAmount] = useState("");
  const [fromCurrency, setFromCurrency] = useState("USD");
  const [toCurrency, setToCurrency] = useState("INR");
  const [result, setResult] = useState("");

  const rates = {
    USD: 1,
    INR: 83,
    EUR: 0.92,
    GBP: 0.79,
  };

  const convertCurrency = () => {
    const amountInUSD = amount / rates[fromCurrency];
    const convertedAmount = amountInUSD * rates[toCurrency];

    setResult(convertedAmount.toFixed(2));
  };

  return (
    <div style={{ textAlign: "center", marginTop: "50px" }}>
      <h1>Currency Converter</h1>

      <input
        type="number"
        placeholder="Enter Amount"
        value={amount}
        onChange={(e) => setAmount(e.target.value)}
      />

      <br /><br />

      <select
        value={fromCurrency}
        onChange={(e) => setFromCurrency(e.target.value)}
      >
        <option value="USD">USD</option>
        <option value="INR">INR</option>
        <option value="EUR">EUR</option>
        <option value="GBP">GBP</option>
      </select>

      <span> → </span>
    </div>
  );
}

```

```

35
36     <select
37       value={fromCurrency}
38       onChange={(e) => setFromCurrency(e.target.value)}
39     >
40       <option value="USD">USD</option>
41       <option value="INR">INR</option>
42       <option value="EUR">EUR</option>
43       <option value="GBP">GBP</option>
44     </select>
45
46     <span> → </span>
47
48     <select
49       value={toCurrency}
50       onChange={(e) => setToCurrency(e.target.value)}
51     >
52       <option value="USD">USD</option>
53       <option value="INR">INR</option>
54       <option value="EUR">EUR</option>
55       <option value="GBP">GBP</option>
56     </select>
57
58     <br /><br />
59
60     <button onClick={convertCurrency}>Convert</button>
61
62     {result && (
63       <h2>
64         |   Converted Amount: {result} {toCurrency}
65       </h2>
66     )}
67   </div>
68 );
69 }
70
71 export default App;

```

6) digital Counter

```
App.jsx M ● localhost:5173 X react X
react > src > App.jsx > default
1  import { useState } from "react";
2
3  function App() {
4    const [count, setCount] = useState(0);
5
6    const increment = () => {
7      | setCount(count + 1);
8    };
9
10   const decrement = () => {
11     | setCount(count - 1);
12   };
13
14   const reset = () => {
15     | setCount(0);
16   };
17
18   return (
19     <div style={{ textAlign: "center", marginTop: "50px" }}>
20       <h1>Digital Counter</h1>
21
22       <h2>{count}</h2>
23
24       <button onClick={increment}>Increment (+)</button>
25
26       <button
27         | onClick={decrement}
28         | style={{ marginLeft: "10px" }}
29       >
30       | Decrement (-)
31     </button>
32
33     <button
34       | onClick={reset}
35       | style={{ marginLeft: "10px" }}
36     >
37     | Reset
38   </button>
39 </div>
40 );
41 }
42
43 export default App;
```

7) expense Calculator

```
App.jsx M • localhost:5173 X react X
react > src > App.jsx > default
1  import { useState } from "react";
2
3  function App() {
4    const [description, setDescription] = useState("");
5    const [amount, setAmount] = useState("");
6    const [transactions, setTransactions] = useState([]);
7
8    const addTransaction = () => {
9      if (description === "" || amount === "") {
10        alert("Please enter all fields");
11        return;
12      }
13
14      const newTransaction = {
15        id: Date.now(),
16        description,
17        amount: Number(amount),
18      };
19
20      setTransactions([...transactions, newTransaction]);
21      setDescription("");
22      setAmount("");
23    };
24
25    const deleteTransaction = (id) => {
26      setTransactions(
27        transactions.filter((item) => item.id !== id)
28      );
29    };
30
31    const total = transactions.reduce(
32      (sum, item) => sum + item.amount,
33      0
34    );
35
36    return (
37      <div style={{ textAlign: "center", marginTop: "30px" }}>
38        <h1>Expense Tracker</h1>
39
40        <input
41          type="text"
42          placeholder="Description"
43          value={description}
44          onChange={(e) => setDescription(e.target.value)}
45        />
46
47        <br /><br />
48
49        <input
50          type="number"
51          placeholder="Amount"
52          value={amount}
```

```

3     value={description}
4     onChange={(e) => setDescription(e.target.value)}
5   />
6
7   <br /><br />
8
9   <input
10     type="number"
11     placeholder="Amount"
12     value={amount}
13     onChange={(e) => setAmount(e.target.value)}
14   />
15
16   <br /><br />
17
18   <button onClick={addTransaction}>
19     Add Transaction
20   </button>
21
22   <h2>Total Expense: ₹{total}</h2>
23
24   <h3>Transactions</h3>
25
26   <ul style={{ listStyle: "none", padding: 0 }}>
27     {transactions.map((item) => (
28       <li key={item.id} style={{ margin: "10px" }}>
29         {item.description} - ₹{item.amount}
30
31         <button
32           onClick={() => deleteTransaction(item.id)}
33           style={{ marginLeft: "10px" }}
34         >
35           Delete
36         </button>
37       </li>
38     ))}
39   </ul>
40 </div>
41 );
42 }
43
44 export default App;

```

8) Student grade calculator

```
App.jsx M  localhost:5173 X  react X
react > src > App.jsx > App > calculateGrade
1  import { useState } from "react";
2  function App() {
3      const [mark1, setMark1] = useState("");
4      const [mark2, setMark2] = useState("");
5      const [mark3, setMark3] = useState("");
6      const [total, setTotal] = useState(null);
7      const [average, setAverage] = useState(null);
8      const [grade, setGrade] = useState("");
9      const calculateGrade = () => {
10         const m1 = Number(mark1);
11         const m2 = Number(mark2);
12         const m3 = Number(mark3);
13         const totalMarks = m1 + m2 + m3;
14         const avg = totalMarks / 3;
15         let studentGrade = "";
16         if (avg >= 90) {
17             studentGrade = "A+";
18         } else if (avg >= 80) {
19             studentGrade = "A";
20         } else if (avg >= 70) {
21             studentGrade = "B";
22         } else if (avg >= 60) {
23             studentGrade = "C";
24         } else if (avg >= 50) {
25             studentGrade = "D";
26         } else {
27             studentGrade = "F";
28         }
29         setTotal(totalMarks);
30         setAverage(avg.toFixed(2));
31         setGrade(studentGrade);
32     };
33
34     const reset = () => {
35         setMark1("");
36         setMark2("");
37         setMark3("");
38         setTotal(null);
39         setAverage(null);
40         setGrade("");
41     };
42
43     return (
44         <div style={{ textAlign: "center", marginTop: "50px" }}>
45             <h1>Student Grade Calculator</h1>
46
47             <input
48                 type="number"
49                 placeholder="Subject 1 Marks"
50                 value={mark1}
51                 onChange={(e) => setMark1(e.target.value)}
52             />
```

